Zoho second round interview qeustions

Textile technology (Anna University)

Question 01

Write a program that takes two integers as input, 'n' and 'm', and generates an "n x m" matrix filled with the elements of the Fibonacci sequence in a spiral order, starting from the top-left corner and moving in a clockwise direction.

Example

Input : n = 3, m = 4

Output:

0	1	1	2
34	55	89	3
21	13	8	5

Question 02

Print the string by ignoring alternate occurrences of any character.

Input : Silence speaks

Output : Silenc peaks

Input: The clock ticked rapidly

Output: The clok c id rapy

Question 05

Imagine a cricket match with n overs, where each over comprises 6 balls. Two batsmen face off, eager to pile up runs for their team. However, there are some interesting twists to the usual rules:

No freebies: Forget wides, no-balls, and byes! Only runs scored off the batsman's bat count.

Strike shift: The batsman facing the bowler changes strike only for runs scored, not for boundaries (fours or sixes). So, hitting a six will not swap the strike, but a single will!

Two's company, wickets none: There are only two batsmen in this match, and wickets do not exist. The batsmen keep batting until the overs are done.

Challenge:

You are given n strings, each containing 6 characters (space separated), each character being integer or a dot(. - No run scored in that ball). Each integer represents the total runs scored for that ball. Calculate and return the number of runs scored by each batsman at the end of the match.

Example:

Input:

No of overs: 3

Overs:

1 2 6 . . 4

4 2 2 1 . .

2 6 4 . 1 1

Output:

P1: 23

P2: 13

Write a program that takes two integers as input 'n' and ...

Question 06

Given an image represented as a 2D matrix where each element (image[i][j]) contains integer values representing the RGB color of a pixel. Implement a function or method to identify distinct color regions in the image. Each color region is formed by adjacent pixels with the same color. The function should then replace the colors of any k regions in the image with specified new colors. The function should return the no of distinct color regions in updated image and updated image with new colors.

Input:

210 210 255 255 255
210 210 255 255 255
255 255 267 255 255
255 255 255 210 210

k=1

newColors: {210: 255}

Output:

No of distinct color portions: 3

Updated Image:

255 255 255 255 255
255 255 255 255 255
255 255 267 255 255
255 255 255 210 210